

WHACK-A-MOLE GAME

Technical Documentation

Name- Antra Agarwal

Roll no -2410110404

I made Whack-A-Mole game which tested my Java and OOPS concepts like Inheritance and Threading . This game basically consists of a grid of holes from which random objects pop up and we get points accordingly whether we click on the Mole or the Bomb to maximize our score within the given time limit. This project taught me the practical implementation of many concepts which are detailed below :-

1)Object-Oriented Design

Abstraction:-All Hole items share some common traits like visibility and remaining lifetime so we use the abstract class HoleOccupant to encapsulate these shared traits like isVisible()

Inheritance:-Mole ,Bomb and BonusMole extends HoleOccupant and they inherit the behaviour as established in HoleOccupant.

Polymorphism:-When we click a button , the system doesn't ask whether it's a mole ,bomb or bonus mole , it just gives the points according to their respective values.

Encapsulation:- Game-state variables are private inside GameEngine.

2)Concurrency Strategy

So that the game keep going on while moles spawn and the timer counts down, we use **multithreading**.

The Game Engine Thread

GameEngine implements Runnable and runs its game loop on a thread in the background

This loop takes care of updating timer, ticking occupants, spawning new occupants and removing expired ones

If we ran this loop on the main thread, the buttons would freeze and we wouldn't be able to click anything. By putting it on a different thread, the "Event Dispatch Thread" stays free to handle clicks.

The engine updates the UI (like the timer label) by calling methods in the main class. The engine never modifies Swing components directly. Whenever the engine needs to change the UI (score, icons, timer), it uses: `SwingUtilities.invokeLater(...)`

3)Exception Handling

InterruptedException

We have wrapped `Thread.sleep(1000)` in a try-catch block in the Main loop. When the user closes the window out of nowhere, the game interrupts the engine thread and the program gets closed down immediately.

HighScoreException (Checked)

The `HighScoreManager` class uses Java serialization to save and load top scores.

InvalidGameStateException (Unchecked)

This custom runtime exception shows those situations that should not happen anytime like trying to spawn new object when there is no free hole